Digital Technologies and Science Education I · Professional Master of Education, Year 1 · 4 November 2026

Integrating Technology into Science Education

The CRAFT pedagogical cycle with artificial intelligence (AI)

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Before step: write your own answer first



Check the Machine

The task

Explain to a 5th class pupil why we have seasons.

- Groups of three: write your own answer first, one minute
- Then we show what an AI wrote

After step: what one AI wrote

Task → Before → **After** → Takeaway

Check the Machine

> “Earth experiences seasons because of its elliptical orbit around the Sun. When Earth is closer to the Sun, we have summer, and when it is farther away, we have winter. That’s why the Northern Hemisphere is warmer in July and colder in January — Earth’s distance to the Sun changes throughout the year.”

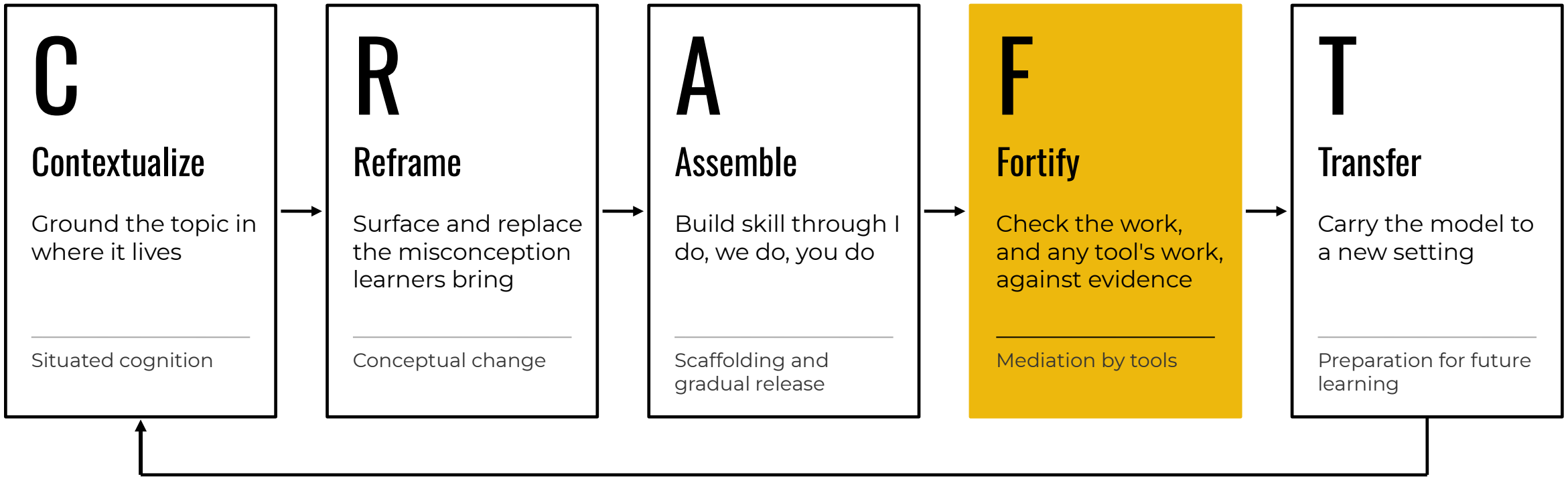
AI-generated explanation, from the AI Error Gallery of the CRAFT PD Series (cxedhub.com)

- Where does it match your group's answer?
- What would you check it against?

CRAFT · Contextualize · Reframe · Assemble · Fortify · Transfer

Build understanding. Trust nothing without evidence.

The CRAFT cycle



One cycle per lesson · AI is one more tool whose output learners check

Contextualize phase: where do seasons live?

Here

About 17 hours of daylight in June, under 8 in December

Elsewhere

Christmas falls in summer in Sydney

In your plan

The curriculum objective the lesson serves

- The lesson is about seasons; the chatbot is one resource among several

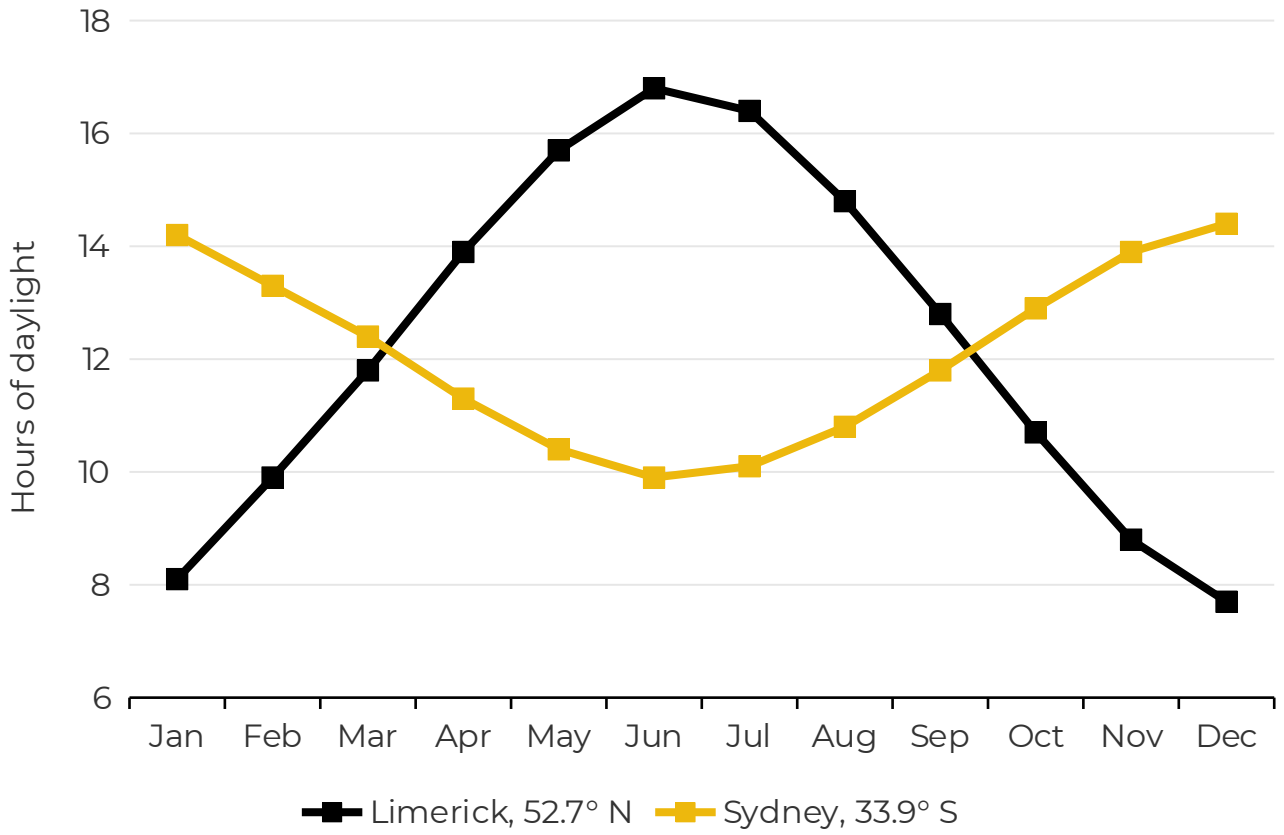
Reframe phase: summer is not when Earth is closest

Belief

Summer comes when Earth is closest to the Sun.

Model

Earth's tilt changes how high the Sun climbs and how long the day lasts.



Calculated for the 15th of each month · Earth is closest to the Sun in early January

Assemble phase: the AI sits inside the worked example

I do: the four parts of our prompt, read aloud as we type it

Role	You are a primary teacher in Ireland planning science for 5th class.
Task	Write a 40-minute lesson on why we have seasons.
Constraints	Use the tilt model, name the distance belief, and include a torch-and-globe activity.
Format	Objective, hook, activities, and one check for understanding.



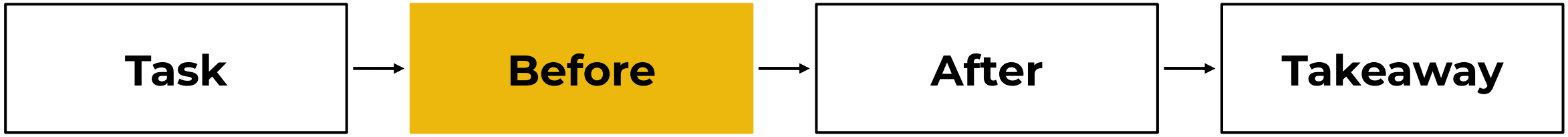
We prompt, read the draft aloud, and reject what fails

Groups of three fill the draft's gaps: its lines for the Reframe and Fortify phases

Plan your own lesson in today's activity

Fortify phase: the Check the Machine routine

Run it in groups of three before anyone runs it alone



What we asked the AI to do

Our answer, written before we look

What the AI produced, set beside ours

What we learned about the tool, and about ourselves

- AI is never the first step
- Errors are features
- The check transfers to any source of answers

Fortify phase: three more ways to check

Ask a second machine

Same prompt, two chatbots. A disagreement is a signal worth investigating.

Check a source

Set each claim against the curriculum, a textbook, or a science reference.

Try to break it

Ask the question a pupil holding the misconception would ask.

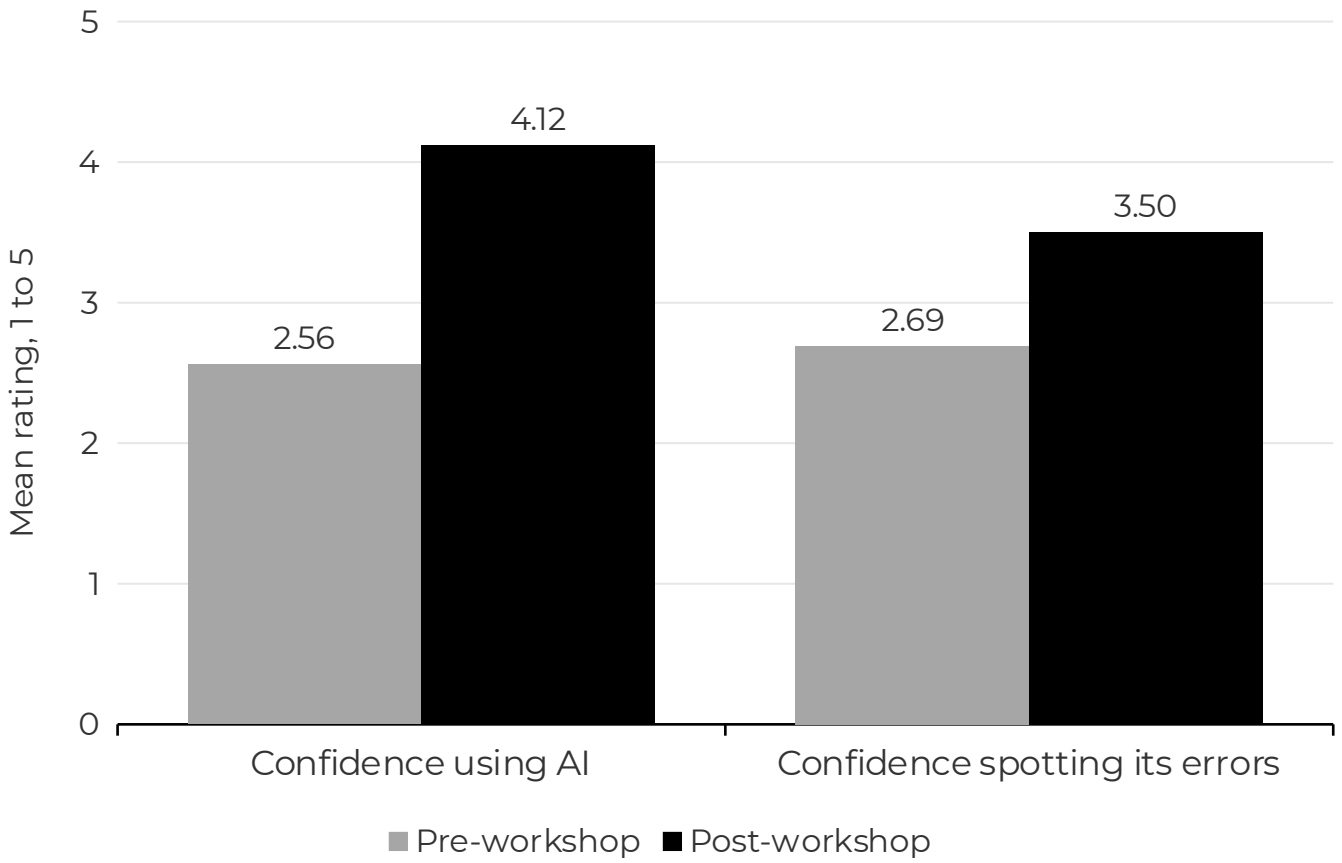
Evidence pupils can make
Shine a torch straight down on squared paper, then at a slant. Count the squares it lights each time.

Fortify phase checks the work; 5E's Evaluate phase checks the learner

Evaluate phase (5E)	Fortify phase (CRAFT)
Did the learner meet the objective?	Does the work, and any tool's output, hold up against evidence?
The teacher judges	Learners check, in groups of three first, then alone
Ends the cycle	Comes before the Evaluate phase and can run as its own small cycle

- What the Fortify phase checks: the lesson plan, the AI's explanation, and the pupils' own claims

Using AI and checking AI grow at different rates



+1.56

using AI

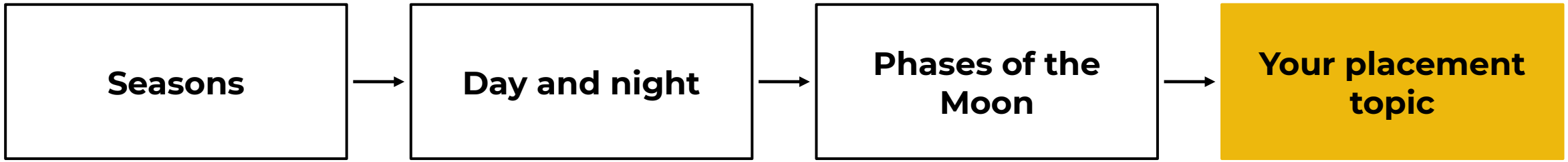
+0.81

spotting its errors

- One session teaches people to use AI; repeated, structured checking teaches them when to trust it

One-day CRAFT workshop for K–12 STEM teachers in a regional Noyce network · matched pre and post, N = 16 · preliminary results, draft manuscript

Transfer phase: same check, next topic



Tilt, not distance

Earth spins; the Sun does not circle us

Angle of view, not Earth's shadow

Name the belief, then run Check the Machine

Every AI answer earns the same question: checked against what?

- With primary pupils, the teacher runs the AI; pupils predict, compare, and record

Your turn: plan one CRAFT lesson with AI

- Groups of three · 15 minutes · one device per group
- Pick a topic and its belief from the menu
- Before step: write what a correct answer must say
- Prompt the AI for a draft, then run the Check the Machine steps
- Mark the draft: Keep, Fix, or Cut



CRAFT lesson planner with AI

Class level _____ Topic _____ AI tool _____

Contextualize phase Where does this live in pupils' lives? Which curriculum objective?	
Reframe phase What belief will pupils bring? What counterexample breaks it?	
Assemble phase Prompt: role · task · constraints · format. I do · we do · you do	
Fortify phase Check the Machine steps: Task · Before · After · Takeaway. Checked against what?	
Transfer phase Where does the check go next?	

AI draft verdict: Keep · Fix · Cut

Topic and belief menu

Topic · the belief pupils bring

- Seasons · summer is when Earth is closest to the Sun
- Day and night · the Sun travels around Earth
- Falling objects · heavier things fall faster
- Plants · plants get their food from the soil
- Materials · metal is colder than wood in the same room
- Sound · sound travels best through air
- Magnets · magnets attract every metal

Check it against

- A second chatbot
- The curriculum or a textbook
- A quick experiment pupils run
- A question that tries to break it

Share-out · 7 minutes

Share what your group wrote at the Takeaway step: what did the machine get wrong, and what did you check it against?

Take it to placement

- Ask pupils for a Before-step answer, then show the machine
- Run one Fortify-phase check in every unit
- Free Check the Machine worksheet and content audit checklist: cxedhub.com

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